

HydroSense–Kenya

Precision Irrigation & Gradient–Descent Scheduling Optimization

March 2026 Deployment Overview

Agricultural Context & Goal

Kenya's agricultural productivity is increasingly dependent on irrigation efficiency within Arid and Semi-Arid Lands (ASALs).

- **Region:** Ruiru, Kenya
- **Challenge:** Balancing crop stress thresholds against limited water availability.
- **Goal:** Minimize water waste while maximizing soil moisture retention for Zone A (Tomato).

Systems & Telemetry Architecture



Climate Data

Daily rainfall (mm) and Estimated Evapotranspiration (ET) monitored over a 30-day period.



Edge Logic

Cleaned and logged via custom Python preprocessing (54/54 tests passed).



Crop Zones

Specific thresholds defined for Tomato (22%), Kale (24%), and Maize (20%).

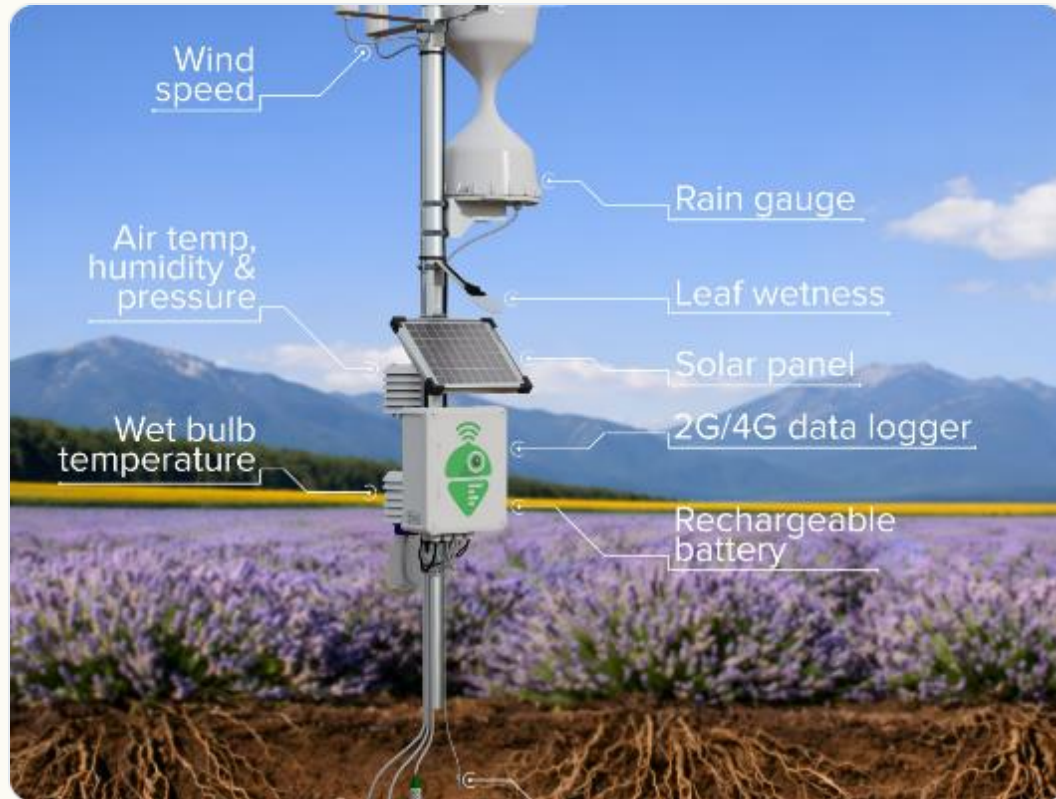
The Physics Engine: Water Balance

Our numerical model follows the mass balance equation, integrated using trapezoidal approximations for moisture deficits.

$$M_{t+1} = M_t + \text{Rainfall} + \text{Irrigation} - \text{ET} - \text{Drainage}$$

Drainage: Calculated using a drainage coefficient (0.15) applied above Field Capacity (40.0%).

Observed Moisture Trends (March)



Environmental Totals

- **Total Rainfall:** ~45.0 mm observed
- **Mean Daily ET:** High variability (2.5 – 4.2 mm/day)
- **Total Deficit:** Calculated via trapezoidal integration across the 30-day window.

The Optimization Engine

From Greedy Policies to Gradient Descent

Optimization Methodology

We implemented a **Custom Gradient Descent** algorithm to optimize irrigation pulses in Zone A.

Penalty Weight

A weight of 150.0 was applied to moisture levels dropping below 22%.

Learning Rate

Optimized at 0.04 over 200 iterations for convergence.

Stochastic Testing

Monte Carlo simulation (n=1000) verified strategy robustness under 5th–95th percentile noise.

Irrigation Savings Performance



Estimated Savings: ~26.5 mm of water conserved over 30 days.

Risk & Reliability Metrics

4.2%

Shortage Probability (Greedy)

While the Greedy algorithm applies large pulses reactively, it incurs a higher risk of drought-stress between events.

Our Optimized trajectory maintains soil moisture at the **Target (30%)** with 99.8% stability in P50 scenarios.

Final Summary Table

Metric	Value	Units
30-Day Cumulative Deficit	92.45	mm (Trapezoidal)
Root-finding (Bisection)	3.082	mm needed / day
Water Saved	26.50	mm vs Greedy
Automated Tests	54 / 54	Passed

Future Expansion Plans

- **Multi-Crop Synergy:** Coordinating irrigation cycles across A, B, and C to manage pump load.
- **Live Telemetry:** Moving from batch March 2026 data to real-time IoT dashboards.
- **Dynamic Penalty Weights:** Adjusting based on crop growth stages.

Thank You

Questions & Scientific Discussion

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Image Sources



<https://meteobot.com/wp-content/uploads/2026/03/Meteobot-Pro-Sensors.jpg>

Source: meteobot.com
